course, is well beyond the realm of our experience. Time, as we know it, is a dimension we can traverse in only one direction with absolute inevitability, never being able to return to an instant after it has passed. Of course, it might be that curled-up time dimensions have vastly different properties from the familiar, vast time dimension that we imagine reaching back to the creation of the universe and forward to the present moment. But, in contrast to extra spatial dimensions, new and previously unknown time dimensions would clearly require an even more monumental restructuring of our intuition. Some theorists have been exploring the possibility of incorporating extra time dimensions into string theory, but as yet the situation is inconclusive. In our discussion of string theory, we will stick to the more "conventional" approach in which all of the curled-up dimensions are space dimensions, but the intriguing possibility of new time dimensions could well play a role in future developments.

The Physical Implications of Extra Dimensions

Years of research, dating hack to Kaluza's original paper, have shown that even though any extra dimensions that physicists propose must be smaller than we or our equipment can directly "see" (since we haven't seen them), they do have important indirect effects on the physics that we observe. In string theory, this connection between the microscopic properties of space and the physics we observe is particularly transparent.

To understand this, you need to recall that masses and charges of particles in string theory are determined by the possible resonant vibrational string patterns. Picture a tiny string as it moves and oscillates, and you will realize that the resonant patterns are influenced by its spatial surroundings. Think, for example, of ocean waves. Out in the grand expanse of the open ocean, isolated wave patterns are relatively free to form and travel this way or that. This is much like the vibrational patterns of a string as it moves through large, extended spatial dimensions. As in Chapter 6, such a string is equally free to oscillate in any of the extended directions at any moment. But if an ocean wave passes through a more cramped spatial environment, the detailed form of its wave motion will surely be affected by, for example, the depth of the water, the placement and shape of the rocks encountered, the canals through which the water is channeled, and so